Why We Don't Discount at the "Risk-Free Rate": Collateral, Repo, and the Post-Crisis Discounting Stack

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Setting and Goal

The article reframes the textbook "discount at the risk-free rate" prescription as a special case of a more general statement: *the discount rate is the funding rate of the replicating strategy.* The single rate r in Black–Scholes is a placeholder collapsing three economically distinct rates that have diverged materially since 2008:

- the *repo rate* r_{repo}, the cost of carrying the hedge in the underlying;
- the *collateral rate* r_c, the rate paid on cash posted under the CSA;
- the *unsecured funding rate* r_f, the dealer's own treasury rate for any uncollateralised exposure.

The goal of these notes is to (i) state the general collateralised-pricing PDE/expectation in a form that connects to the Pucci/Lou framework already in our reference set, (ii) make the three collateral regimes (perfect CSA, no CSA, partial CSA) precise, and (iii) translate the verbal arguments into a checklist of validation tests for a multi-curve / CSA-aware pricing library.

1 General Pricing Framework

1.1 Stochastic-rate formulation

Let V be a derivative on an underlying S, traded between a dealer and a counterparty with a CSA specifying a stochastic collateral rate process $r_c(t)$. Let $r_{\text{repo}}(t)$ be the instantaneous repo rate of the hedge instrument and $r_f(t)$ the dealer's instantaneous unsecured funding rate. Under the pricing measure $\mathbb{Q}$:

Proposition 1 (General collateralised pricing formula)**.** *For a fully collateralised derivative,*

$$V(t) \;=\; \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_c(s)\,ds\right) V(T) \,\middle|\, \mathcal{F}_t\right]. \tag{1}$$

The underlying obeys

$$dS(t) \;=\; r_{\text{repo}}(t)\,S(t)\,dt \;+\; \sigma(t,S(t))\,S(t)\,dW^{\mathbb{Q}}(t), \tag{2}$$

i.e. $\exp(-\int_0^t r_{\text{repo}}(s)\,ds)\,S(t)$ *is a* $\mathbb{Q}$*-martingale,* not $\exp(-\int_0^t r_c(s)\,ds)\,S(t)$*. The repo rate enters the drift; the collateral rate enters the discount factor.*

For an uncollateralised trade replace r_c by r_f in (1). For a partial CSA with threshold H, the discount rate is the path-dependent process

$$r_{\text{eff}}(s) \;=\; r_c(s)\,\mathbf{1}_{\{|V(s)|>H\}} \;+\; r_f(s)\,\mathbf{1}_{\{|V(s)|\le H\}}. \tag{3}$$

1.2 XVA decomposition

For an uncollateralised trade priced off the OIS clean value V_{OIS}, the value to the dealer is

$$V_{\text{uncoll}}(t) \;=\; V_{\text{OIS}}(t) \;-\; \text{CVA}(t) \;+\; \text{DVA}(t) \;-\; \text{FVA}(t), \tag{4}$$

with

$$\text{CVA}(t) = \mathbb{E}^{\mathbb{Q}}\left[\int_t^T e^{-\int_t^s r_c(u)\,du}\,\lambda_C(s)\,(1-R_C)\,V(s)^+\,ds \,\middle|\, \mathcal{F}_t\right], \tag{5}$$

$$\text{DVA}(t) = \mathbb{E}^{\mathbb{Q}}\left[\int_t^T e^{-\int_t^s r_c(u)\,du}\,\lambda_D(s)\,(1-R_D)\,(-V(s))^+\,ds \,\middle|\, \mathcal{F}_t\right], \tag{6}$$

$$\text{FVA}(t) = \mathbb{E}^{\mathbb{Q}}\left[\int_t^T e^{-\int_t^s r_c(u)\,du}\,\big(r_f(s)-r_c(s)\big)\,V(s)\,ds \,\middle|\, \mathcal{F}_t\right]. \tag{7}$$

Here λ_C, λ_D are the stochastic default intensities of the counterparty and dealer; R_C, R_D are recovery rates.

Remark 1 (Constant-rate convention)**.** *For the remainder of the paper, all rates are treated as constants. Each formula generalises by replacing $r\,(T-t)$ with $\int_t^T r(s)\,ds$. The constant-rate formulas suffice to expose the economic logic that distinguishes the three collateral regimes.*

2 The Textbook Argument and Where It Hides Its Assumption

The Black–Scholes derivation: hold V, hedge with $-\Delta = -\partial_S V$ units of S, apply Itô's lemma; the Brownian term cancels; by no-arbitrage the riskless portfolio earns the short rate r, yielding

$$V(t) \;=\; e^{-r(T-t)}\,\mathbb{E}^{\mathbb{Q}}[V(T)\,|\,\mathcal{F}_t]\,. \tag{8}$$

Remark 2 (The hidden assumption)**.** *Equation* (8) *assumes a* single *rate r at which every market participant borrows, lends, funds, and earns collateral. Pre-2008, LIBOR proxied this for everyone within a handful of basis points. Post-2008, the LIBOR–OIS spread blew out to 350bp and the assumption broke. The mathematics did not change: the r in* (8) *is the funding rate of the replicating portfolio, and that rate now depends on the collateral arrangement.*

3 The Repo Rate: Drift, Not Discount

Redoing the hedging argument with care produces the modified PDE

$$\boxed{\frac{\partial V}{\partial t} \;+\; r_{\text{repo}}\,S\,\frac{\partial V}{\partial S} \;+\; \tfrac{1}{2}\sigma^2 S^2\,\frac{\partial^2 V}{\partial S^2} \;=\; r_c\,V.} \tag{9}$$

The repo rate appears in the drift (cost of carry of the hedge); the collateral rate appears on the right (discounting). They are *always* different.

3.1 Equity repo: forward price under cost of carry

For a non-dividend-paying stock with equity repo rate r_{repo},

$$F(t,T) \;=\; S(t)\,e^{r_{\text{repo}}(T-t)}. \tag{10}$$

Worked example (paper, §3.1): $S = £100$, $T = 1\text{y}$, SONIA $= 5.00\%$, $r_{\text{repo}} = 5.50\%$. Then

$$F = 100 \cdot e^{0.055} \approx £105.65, \qquad F_{\text{textbook}} = 100 \cdot e^{0.05} \approx £105.13.$$

A 52p discrepancy per share, i.e. £52,000 on a 100k-share option, before any volatility effect. If the stock becomes hard to borrow ($r_{\text{repo}} \to 8\%$), $F \to £108.33$.

3.2 Bond repo: GC vs special

For a coupon-bearing bond with continuously compounded coupon yield c and bond repo rate $r_{\text{repo}}^{\text{bond}}$,

$$F^{\text{bond}}(t,T) \;=\; P(t)\, e^{(r_{\text{repo}}^{\text{bond}} - c)(T-t)}. \tag{11}$$

Specialness shifts the forward and propagates into swaption hedge costs. The paper's §3.2 example: a £50m 10-year benchmark Gilt with duration ≈ 8.5, SONIA at 5.00%, special repo at 3.50% gives

$$\Delta\text{PV} \;\approx\; (\text{SONIA} - r_{\text{repo}}^{\text{special}}) \;\times\; N \;\times\; \text{Dur} \;=\; 1.50\% \times £50\text{m} \times 8.5 \;\approx\; £6.4\text{m}.$$

3.3 Repo vs collateral – the structural distinction

	Repo rate r_{repo}	**Collateral rate** r_c
What it is	Cost of borrowing the hedge asset	Rate earned on posted margin
Where it enters	Drift (forward price of the hedge)	Discounting (PV of future cash flows)
Determined by	Repo market for each asset	CSA contract between counterparties
Can they differ?	Yes – always	Yes – always

4 The Three Collateral Regimes

Definition 1 (Collateral regimes)**.** *A bilateral derivatives trade falls into one of three regimes according to the CSA in force:*

1. ***Regime 1 – Perfect CSA:*** *zero threshold, daily cash margining in a single currency. Collateral $C(t) = V(t)$ at all t.*
2. ***Regime 2 – No CSA:*** *no collateral. Exposure equal to $V(t)$ for the lifetime of the trade.*
3. ***Regime 3 – Partial CSA:*** *non-trivial threshold $H > 0$, minimum transfer amount, or non-cash collateral eligibility.*

4.1 Regime 1: perfect CSA

Proposition 2 (Perfect-CSA discount rate)**.** *Under a perfect CSA with collateral rate r_c on cash collateral, the no-arbitrage value is*

$$\boxed{V(t) \;=\; e^{-r_c(T-t)}\, \mathbb{E}^{\mathbb{Q}}[V(T) \,|\, \mathcal{F}_t]\,.} \tag{12}$$

Sketch. The dealer holds $V(t)$, receives cash $C(t) = V(t)$ from the counterparty, pays $r_c\, C(t)\, dt$ on the collateral and invests the cash overnight at r_c, so the net carry on the collateral leg is zero. For the discounted value $e^{-r_c t}V(t)$ to be a $\mathbb{Q}$-martingale (no-arbitrage), V must drift at $r_c V$. Discounting at any other rate $\tilde{r} \neq r_c$ produces a deterministic drift $(r_c - \tilde{r})V(t)\, dt$ in $e^{-\tilde{r}t}V(t)$ – an arbitrage. □

In practice r_c is the overnight rate of the CSA currency (SOFR, SONIA, STR) and the discount curve is bootstrapped from OIS swaps referencing it.

4.2 Regime 2: no CSA

Proposition 3 (Uncollateralised discount rate)**.** *For an uncollateralised trade, the dealer's funding rate r_f replaces r_c:*

$$V(t) \;=\; e^{-r_f(T-t)}\,\mathbb{E}^{\mathbb{Q}}[V(T)\,|\,\mathcal{F}_t]\,, \tag{13}$$

and the law of one price breaks: a AA-rated bank and an A–rated bank arrive at different prices for the same cash flows. Desks restore a common reference by pricing on OIS and adding XVA terms (4)*.*

4.3 Regime 3: partial CSA

Proposition 4 (Partial-CSA effective rate)**.** *With threshold H, MTAs, or non-cash collateral, the effective discount rate is the path-dependent process* (3)*, and*

$$V(t) \;=\; \mathbb{E}^{\mathbb{Q}}\left[\exp\left(-\int_t^T r_{\text{eff}}(s)\,ds\right) V(T)\,\middle|\,\mathcal{F}_t\right]. \tag{14}$$

The expectation no longer factorises: the discount rate depends on the realised exposure, so even with constant base rates the valuation requires Monte Carlo simulation of the joint exposure / collateral-state dynamics.

5 Worked Example: 5y USD Receiver Swap

Regime	Discount rate	Rate	PV (Fixed Leg)	Impact vs OIS
Perfect CSA	r_c (SOFR)	4.00%	\$15.54m	base
No CSA	r_f (dealer fdg)	4.80%	\$14.97m	−\$570k
Partial (\$10m thr.)	r_{eff} (blended)	≈ 4.35%	\$15.29m	−\$250k

Table 1: Section 6.3 of the paper. 5y receiver swap, fixed leg 3.50%, notional \$100m. The \$570k gap on a vanilla swap is the order of magnitude that drives competitive pricing.

6 Cross-Currency: CSA Currency Matters

For a EUR/USD trade where Dealer A posts EUR collateral at r_c^{EUR} and Dealer B posts USD at r_c^{USD},

$$V^A(t) = e^{-r_c^{\text{EUR}}(T-t)}\,\mathbb{E}^{\mathbb{Q}}[V(T)\,|\,\mathcal{F}_t]\,, \tag{15}$$

$$V^B(t) = e^{-r_c^{\text{USD}}(T-t)}\,\mathbb{E}^{\mathbb{Q}}[V(T)\,|\,\mathcal{F}_t]\,. \tag{16}$$

The two values differ because the funding rates of the two collateralisation strategies differ. Cross-currency basis swaps quote this differential.

Remark 3 (Cheapest-to-deliver collateral)**.** *A multi-currency CSA with several eligible collateral currencies embeds an optionality: the posting party will deliver in the currency whose collateral rate is cheapest to fund. The bank receiving the collateral effectively earns the* minimum *of the eligible collateral rates. Modelling this requires a basket-min optionality on the collateral side – structurally similar to the cheapest-to-deliver in a futures-style margin scheme.*

7 Non-Cash Collateral: Repo Enters the Discount Rate

When the CSA permits delivery of bonds in lieu of cash, the receiving party's effective collateral return is the repo rate of the posted bond, not the cash CSA rate. The bank monetises the bond via repo:

Cash flow	**Rate**
Cash from repo, invested overnight	$+r_c$ (CSA rate)
Repo paid on the posted bond	$-r_{\text{repo}}^{\text{bond}}$
Net effective collateral return	$r_c - (r_c - r_{\text{repo}}^{\text{bond}}) = r_{\text{repo}}^{\text{bond}}$

Proposition 5 (Effective discount rate with bond collateral)**.** *For a trade collateralised by a specific bond with repo rate $r_{\text{repo}}^{\text{bond}}$ and haircut cost h,*

$$\boxed{r_{\text{eff}} \;=\; r_{\text{repo}}^{\text{bond}} \;+\; h.} \tag{17}$$

The discount rate is no longer the CSA overnight rate – it is the repo rate of the posted collateral, which varies by bond, by date, and by market conditions.

7.1 ColVA: collateral basis valuation adjustment

The difference between the CSA-cash regime and the bond-collateral regime is captured as a valuation adjustment. For a swap of duration Dur and notional N,

$$\text{ColVA} \;\approx\; (r_c - r_{\text{repo}}^{\text{bond}}) \cdot N \cdot \text{Dur}. \tag{18}$$

GC scenario. Gilt at GC repo 4.95% vs SONIA 5.00%, $N = £20\text{m}$, Dur ≈ 8.5:

$$\text{ColVA} \;\approx\; 0.05\% \times £20\text{m} \times 8.5 \;\approx\; £85{,}000.$$

Special scenario. Gilt trading special at 3.50%:

$$\text{ColVA} \;\approx\; 1.50\% \times £20\text{m} \times 8.5 \;\approx\; £2.55\text{m}.$$

Remark 4 (CTD optionality)**.** *With a basket of eligible bonds, the posting counterparty has a cheapest-to-deliver option: post the bond trading most special (earning the receiver the least). Quantitatively this means the receiver earns the minimum repo rate over the eligible set:*

$$r_c^{\text{eff,CTD}}(t) \;=\; \min_{i \in \mathcal{E}(t)} \big(r_{\text{repo}}^{(i)}(t) + h^{(i)}\big), \tag{19}$$

where $\mathcal{E}(t)$ is the eligible set at time t.

7.2 Two distinct repo channels

The repo rate enters derivatives pricing in two structurally distinct places:

1. *Drift channel* – the repo rate of the *hedge instrument* appears in the drift of the underlying under $\mathbb{Q}$ (Equation (9)).
2. *Discount channel* – the repo rate of the *posted collateral bond* appears in the discount factor when the CSA permits non-cash collateral (Equation (17)).

These involve different bonds with different repo rates on different days; conflating them leads to mispricing.

8 The Discount-Rate Decision Tree

1. **Cleared trade?** Discount at the CCP's overnight rate (this is a perfect CSA by construction, with the CCP as guarantor).
2. **Bilateral with CSA?**
   - Single currency, cash collateral $\Rightarrow$ discount at r_c;
   - Multi-currency or non-cash collateral $\Rightarrow$ discount at OIS, compute ColVA from (17)–(18).
3. **Uncollateralised?** Discount at OIS for the clean V_{OIS}, then layer CVA, DVA, FVA via (4).

9 From Single-Curve to Multi-Curve

Pre-crisis: a single LIBOR curve projected forwards *and* discounted cash flows. Post-crisis:

- the discount curve is built from OIS instruments referencing the CSA rate;
- one or more projection curves estimate future LIBOR / IBOR / RFR fixings (each tenor is a separate curve);
- curves are bootstrapped *jointly* – OIS first, then projection curves stripped using OIS discounting on the swap PV;
- convexity adjustments depend on the correlation between projection and discount curves;
- cross-currency swaps require three or four curves (domestic OIS, foreign OIS, projection curves on each side, plus the FX basis adjustment to align the CSA currency).

Remark 5 (Vanilla swap valuation depends on CSA currency)**.** *Even a single-currency vanilla swap has a different fair value under EUR-collateralised and USD-collateralised CSAs (Equations* (15)–(16)*). This is a post-multi-curve concept that does not exist in the pre-crisis single-curve world.*

10 The Correct Mental Model

Proposition 6 (Discounting decomposition)**.** *The pricing rate is determined by three economic quantities:*

$$\boxed{\begin{array}{lcl} \textit{drift under } \mathbb{Q} & : & r_{\text{repo}} \quad \textit{(cost of carry of the hedge)}, \\ \textit{discount factor} & : & r_c \quad \textit{(rate paid to the collateral)}, \\ \textit{XVA wedge} & : & r_f - r_c \quad \textit{(uncollateralised funding spread, via FVA)}. \end{array}} \tag{20}$$

In the pre-crisis world, the three were close enough to treat as one. Post-crisis they diverge materially, and the pricing infrastructure (multi-curve bootstrapping, CSA-aware valuation engines, XVA Monte Carlo) exists precisely to track the wedges.

11 Connection to the Other Papers in the Reference Set

Lou (TRS, SSRN 3217420). This paper's regime taxonomy is exactly the framework Lou applies trade-by-trade for total return swaps. Lou's switching effective rate r_w is a special case of (3) where the switching condition tracks not the threshold but the sign of the exposure for an uncollateralised TRS, or the repo-style margin for a customised CSA. Under full CSA, Lou's result that "TRS shall remain to be discounted at the cash deposit rate, e.g. the OIS curve" is a direct corollary of Proposition 2.

Pucci (CMS / CMASW / CMT, SSRN 1961545, 2056277, 3387961). The Pucci convexity-correction calculations are performed in the swaption / annuity / T-forward measure – numeraires built from *discount bonds* which are themselves the perfectly-collateralised zero-coupon bond prices implied by Proposition 2. The OIS discount curve underlies the entire convexity machinery. The credit-risk-adjusted discount $\widehat{\widehat{D}}$ used in Pucci's CMT paper is the Regime-2 / unsecured analogue of the OIS discount D.

Pucci T-Lock (SSRN 3386521). The forward bond price formula uses the bond repo rate as the carry, exactly the *drift channel* of Section 3 of these notes.

In short: the present paper is the framework paper underlying the discounting conventions assumed (often implicitly) by every other paper in this collection.

12 Validation Notes for the Pricing Library

The targeted validation exercises below are organised so each test isolates a single feature of the multi-curve / CSA stack.

12.1 Single-curve vs OIS regression

1. **Flat-curve sanity.** Set $r_c = r_f = r_{\text{repo}}$ to a single constant. The library should reproduce the textbook Black–Scholes price for a vanilla call / put. Use this as the regression test for the implementation of Equation (8).
2. **LIBOR-discount legacy mode.** If the library has a legacy single-curve mode, price a 5y vanilla swap with both single-curve and multi-curve modes; the difference should match the LIBOR–OIS spread integrated over the swap duration to within basis-point tolerance.

12.2 Repo drift channel (Equation 9)

3. **Equity forward under repo.** For a non-dividend stock, verify $F(0,T) = S_0\, e^{r_{\text{repo}}T}$ when $r_{\text{repo}} \neq r_c$. Replicate the paper's example: $S_0 = 100$, $r_{\text{repo}} = 5.5\%$, $T = 1\text{y} \Rightarrow F = 105.65$ (vs 105.13 at $r_c = 5\%$).
4. **Vanilla call price under two rates.** Use Black 76 on the forward $F = S_0 e^{r_{\text{repo}}T}$ and discount the option payoff at r_c. Set $S_0 = 100$, $K = 100$, $\sigma = 20\%$, $T = 1$, $r_{\text{repo}} = 5.5\%$, $r_c = 5.0\%$. The library should agree with the closed form

$$C \;=\; e^{-r_c T}\big(F\,N(d_1) - K\,N(d_2)\big), \quad d_{1,2} = \frac{\ln(F/K) \pm \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}.$$

5. **Bond forward with specialness.** Reproduce (11) for a coupon bond. Test GC ($r_{\text{repo}} = 4.95\%$) and special ($r_{\text{repo}} = 3.50\%$) and confirm the forward price difference matches $P_0(e^{(r^{GC}_{\text{repo}}-c)T} - e^{(r^{\text{special}}_{\text{repo}}-c)T})$.

12.3 Collateral discount channel (Equation 12)

6. **Three-regime swap PV.** Replicate the paper's worked example (§6.3): 5y receiver swap, fixed 3.50%, notional \$100m, floating leg = SOFR. Build OIS curve at 4.00% flat. Verify:
   - Perfect-CSA PV $\approx$ \$15.54m on the fixed leg;
   - No-CSA PV at $r_f = 4.80\% \approx$ \$14.97m ($-$\$570k);
   - Partial-CSA PV at $r_{\text{eff}} \approx 4.35\% \approx$ \$15.29m ($-$\$250k).

The differences must be reproduced to a few thousand dollars; this is the tightest discounting test in the paper.

7. **Martingale check.** For the perfect-CSA case, verify that $e^{-r_c t}V(t)$ is a $\mathbb{Q}$-martingale by Monte Carlo simulating $V(t)$ under the deal's hedging dynamics. Check that the discounted process drifts within Monte Carlo error of zero.

12.4 Cross-currency CSA

8. **EUR vs USD CSA on the same trade.** Price a 10y EUR/USD cross-currency swap under (a) USD-cash CSA, (b) EUR-cash CSA. The two PVs must differ; the difference should equal the integral of the cross-currency basis over the swap, discounted at the relevant collateral rate. This validates Equations (15)–(16).

9. **Cheapest-to-deliver CSA.** For a multi-currency CSA with {USD, EUR, JPY} collateral eligibility, price under the CTD assumption (19) and check the PV is the minimum over the three single-currency CSA prices (or below, if the basket optionality is modelled stochastically).

12.5 Non-cash collateral and ColVA

10. **ColVA, GC scenario.** Reproduce ColVA $\approx$ £85k for a 10y £20m swap with Gilt collateral at GC repo 4.95% vs SONIA 5.00% (Equation (18), Dur $\approx$ 8.5).

11. **ColVA, special scenario.** Same trade with a Gilt at special repo 3.50% should give ColVA $\approx$ £2.55m. Validates the magnitude of the collateral-basis effect.

12. **ColVA decomposition.** For each, confirm the library splits the PV as $V_{\text{OIS}} - \text{ColVA}$ and that the perfect-CSA mode equals V_{OIS} exactly.

12.6 Partial CSA / threshold

13. **Path-dependence test.** For a partial CSA with threshold H, simulate the exposure $V(s)$, apply the switching discount rate (3), and verify that $H \to 0$ recovers the perfect-CSA PV and $H \to \infty$ recovers the no-CSA PV. This is a smoke test on the threshold logic.

14. **Linear interpolation falsification.** A common shortcut is to interpolate r_{eff} linearly between r_c and r_f as a function of average exposure. Verify the library does *not* use this shortcut: it should agree with a Monte Carlo path-by-path simulation, not a deterministic blend. The two approaches diverge when the exposure profile has high gamma w.r.t. rates.

12.7 XVA stack

15. **No-CSA $\Rightarrow$ XVA additivity.** For an uncollateralised vanilla swap, verify the decomposition (4): price the trade in (a) "OIS clean" mode and (b) "no CSA, full XVA" mode; the difference must equal $-\text{CVA} + \text{DVA} - \text{FVA}$ as computed independently from (5)–(7).

16. **Symmetry check.** For two counterparties of identical credit and funding profiles entering opposite sides of the trade, $\text{CVA}_A = \text{DVA}_B$ and vice versa. The total XVA wedge should sum to zero across the two desks (modulo the funding-cost asymmetry captured by FVA's first-order direction).

17. **FVA sign.** For an asset-positive trade, $V > 0$ and $r_f > r_c$ together imply FVA > 0 (a cost for the dealer). Falsify the library if it produces a negative FVA on this leg.

12.8 Curve construction

18. **OIS bootstrap regression.** Build the OIS discount curve from a standard set of OIS swaps (1m, 3m, 6m, 1y, 2y, 5y, 10y, 30y). Reprice each input swap with the bootstrapped curve; PV must be zero to machine precision.

19. **LIBOR / IBOR projection curve.** Bootstrap the IBOR projection curve using OIS discounting (*not* self-discounted). Reprice the 3m IBOR swaps used as inputs and verify PV = 0. Then price a 6m-IBOR swap with the 6m projection curve; it should not equal the 3m result (this is the multi-curve fingerprint).

20. **Convexity adjustment.** For a CMS swap, verify the convexity adjustment depends on the correlation between the projection curve and the OIS discount curve. Setting correlation to zero should match a closed-form "no-convexity" approximation; non-zero correlation produces a measurable shift in line with the Pucci CMS framework.

12.9 Numerical stability

21. $r_c \to r_f$ **limit.** As the spread $r_f - r_c$ collapses to zero, both the perfect-CSA and uncollateralised PVs must converge, and the FVA term must vanish. A non-zero residual at zero spread indicates a stale curve or wrong discount factor in the FVA integrand of (7).

22. **Stochastic-rate consistency.** Run two passes: one with rates as constants (the convention of these notes), one with the Hull–White / G2++ short-rate model giving the same expectation $\mathbb{E}[\int_t^T r_c(s)\,ds]$. The two prices should agree at first order; the second-order discrepancy is the convexity adjustment and should match the closed-form Pucci value where applicable.

13 Notation Cheat-Sheet

Symbol	Meaning
r_c	Collateral rate (CSA-specified, overnight: SOFR / SONIA / STR)
r_f	Dealer's unsecured funding rate
r_{repo}	Repo rate of a specific asset (asset-dependent, day-dependent)
r_{eff}	Effective discount rate – mixture of r_c and r_f under partial CSA, or $r_{\text{repo}}^{\text{bond}} + h$ under non-cash collateral
V_{OIS}	Clean derivative value computed with OIS discounting, before XVA
h	Haircut cost on posted collateral
H	CSA threshold (collateral exchange triggers above H)
Dur	Modified duration of the trade
λ_C, λ_D	Stochastic default intensities (counterparty / dealer)
R_C, R_D	Recovery rates (counterparty / dealer)

14 Bibliographic Cross-References

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